

0707.4470 — formula report

213 inline formulas (section omitted; all rendered), 62 display equations, 0 tables, 12 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0025	017	■0.382 ●W +1	$\begin{aligned} & F=E_{\mathrm{x}}\,\mathrm{d}x\wedge\mathrm{d}t+E_{\mathrm{y}}\,\mathrm{d}y\wedge\mathrm{d}t+E_{\mathrm{z}}\,\mathrm{d}z\wedge\mathrm{d}t \\ & \quad +B_{\mathrm{x}}\,\mathrm{d}y\wedge\mathrm{d}z+B_{\mathrm{y}}\,\mathrm{d}z\wedge\mathrm{d}x+B_{\mathrm{z}}\,\mathrm{d}x\wedge\mathrm{d}y, \end{aligned}$	$F = E_x \, \mathrm{d}x \wedge \mathrm{d}t + E_y \, \mathrm{d}y \wedge \mathrm{d}t + E_z \, \mathrm{d}z \wedge \mathrm{d}t \\ + B_x \, \mathrm{d}y \wedge \mathrm{d}z + B_y \, \mathrm{d}z \wedge \mathrm{d}x + B_z \, \mathrm{d}x \wedge \mathrm{d}y$	$F = E_x \, \mathrm{d}x \wedge \mathrm{d}t + E_y \, \mathrm{d}y \wedge \mathrm{d}t + E_z \, \mathrm{d}z \wedge \mathrm{d}t \\ + B_x \, \mathrm{d}y \wedge \mathrm{d}z + B_y \, \mathrm{d}z \wedge \mathrm{d}x + B_z \, \mathrm{d}x \wedge \mathrm{d}y,$
0707.4470_EQ0032	019	■0.582 ●C +40	$\begin{aligned} & \frac{\left(B_{\mathrm{x}}\right)_{\left\{k, l+\frac{1}{2}, m+\frac{1}{2}\right\}}^{n+1}-\left(B_{\mathrm{x}}\right)_{\left\{k, l+\frac{1}{2}, m+\frac{1}{2}\right\}}^n}{\Delta t} = \\ & \frac{\left(E_{\mathrm{y}}\right)_{\left\{k, l+\frac{1}{2}, m+1\right\}}^{n+\frac{1}{2}}-\left(E_{\mathrm{y}}\right)_{\left\{k, l+\frac{1}{2}, m\right\}}^{n+\frac{1}{2}}}{\Delta z}-\frac{\left(E_{\mathrm{z}}\right)_{\left\{k, l+1, m+\frac{1}{2}\right\}}^{n+\frac{1}{2}}-\left(E_{\mathrm{z}}\right)_{\left\{k, l, m+\frac{1}{2}\right\}}^{n+\frac{1}{2}}}{\Delta y} \\ & \frac{\left(B_{\mathrm{y}}\right)_{\left\{k+\frac{1}{2}, l, m+\frac{1}{2}\right\}}^{n+1}-\left(B_{\mathrm{y}}\right)_{\left\{k+\frac{1}{2}, l, m+\frac{1}{2}\right\}}^n}{\Delta t} = \\ & \frac{\left(E_{\mathrm{z}}\right)_{\left\{k+1, l, m+\frac{1}{2}\right\}}^{n+\frac{1}{2}}-\left(E_{\mathrm{z}}\right)_{\left\{k, l, m+\frac{1}{2}\right\}}^{n+\frac{1}{2}}}{\Delta x}-\frac{\left(E_{\mathrm{x}}\right)_{\left\{k+\frac{1}{2}, l, m+1\right\}}^{n+\frac{1}{2}}-\left(E_{\mathrm{x}}\right)_{\left\{k+\frac{1}{2}, l, m\right\}}^{n+\frac{1}{2}}}{\Delta z} \\ & \frac{\left(B_{\mathrm{z}}\right)_{\left\{k+\frac{1}{2}, l+\frac{1}{2}, m\right\}}^{n+1}-\left(B_{\mathrm{z}}\right)_{\left\{k+\frac{1}{2}, l+\frac{1}{2}, m\right\}}^n}{\Delta t} = \\ & \frac{\left(E_{\mathrm{x}}\right)_{\left\{k+\frac{1}{2}, l+1, m\right\}}^{n+\frac{1}{2}}-\left(E_{\mathrm{x}}\right)_{\left\{k+\frac{1}{2}, l, m\right\}}^{n+\frac{1}{2}}}{\Delta y}-\frac{\left(E_{\mathrm{y}}\right)_{\left\{k+1, l+\frac{1}{2}, m\right\}}^{n+\frac{1}{2}}-\left(E_{\mathrm{y}}\right)_{\left\{k, l+\frac{1}{2}, m\right\}}^{n+\frac{1}{2}}}{\Delta x} \end{aligned}$	$\frac{B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1}-B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} =$ $\frac{E_y _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}}-E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z}-\frac{E_z _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}}-E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y}$ $\frac{B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1}-B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} =$ $\frac{E_z _{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}}-E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x}-\frac{E_x _{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}}-E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z}$ $\frac{B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^{n+1}-B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^n}{\Delta t} =$ $\frac{E_x _{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}}-E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y}-\frac{E_y _{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}}-E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x}$	$\frac{B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1}-B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}^n}{\Delta t} =$ $\frac{E_y _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}}-E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta z}-\frac{E_z _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}}-E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y}$ $\frac{B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^{n+1}-B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}^n}{\Delta t} =$ $\frac{E_z _{k+1,l,m+\frac{1}{2}}^{n+\frac{1}{2}}-E_z _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x}-\frac{E_x _{k+\frac{1}{2},l,m+1}^{n+\frac{1}{2}}-E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta z}$ $\frac{B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^{n+1}-B_z _{k+\frac{1}{2},l+\frac{1}{2},m}^n}{\Delta t} =$ $\frac{E_x _{k+\frac{1}{2},l+1,m}^{n+\frac{1}{2}}-E_x _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}}{\Delta y}-\frac{E_y _{k+1,l+\frac{1}{2},m}^{n+\frac{1}{2}}-E_y _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}}}{\Delta x}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0035	020	0.705 C +22	$\begin{aligned} & \begin{aligned} & \frac{\left(D_x \right _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}} - D_x \left _{k+\frac{1}{2},l,m}^{n-\frac{1}{2}} \right)}{\Delta t} = \\ & \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k+\frac{1}{2},l-\frac{1}{2},m}^n \right.}{\Delta y} - \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k+\frac{1}{2},l,m-\frac{1}{2}}^n \right.}{\Delta z} - J_x \left _{k+\frac{1}{2},l,m}^n \right. \\ & \frac{D_y \left _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}} - D_y \left _{k,l+\frac{1}{2},m}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l+\frac{1}{2},m-\frac{1}{2}}^n \right.}{\Delta z} - \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k-\frac{1}{2},l+\frac{1}{2},m}^n \right.}{\Delta x} - J_y \left _{k,l+\frac{1}{2},m}^n \right. \\ & \frac{D_z \left _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}} - D_z \left _{k,l,m+\frac{1}{2}}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k-\frac{1}{2},l,m+\frac{1}{2}}^n \right.}{\Delta x} - \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l-\frac{1}{2},m+\frac{1}{2}}^n \right.}{\Delta y} - J_z \left _{k,l,m+\frac{1}{2}}^n \right. \end{aligned} \end{aligned}$	$\begin{aligned} & \frac{D_x \left _{k+\frac{1}{2},l,m}^{n+\frac{1}{2}} - D_x \left _{k+\frac{1}{2},l,m}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k+\frac{1}{2},l-\frac{1}{2},m}^n \right.}{\Delta y} - \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k+\frac{1}{2},l,m-\frac{1}{2}}^n \right.}{\Delta z} - J_x \left _{k+\frac{1}{2},l,m}^n \right. \\ & \frac{D_y \left _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}} - D_y \left _{k,l+\frac{1}{2},m}^{n-\frac{1}{2}} \right.}{\Delta t} = \\ & \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l+\frac{1}{2},m-\frac{1}{2}}^n \right.}{\Delta z} - \frac{H_z \left _{k+\frac{1}{2},l+\frac{1}{2},m}^n - H_z \left _{k-\frac{1}{2},l+\frac{1}{2},m}^n \right.}{\Delta x} = J_y \left _{k,l+\frac{1}{2},m}^n \right. \\ & \frac{H_y \left _{k+\frac{1}{2},l,m+\frac{1}{2}}^n - H_y \left _{k-\frac{1}{2},l,m+\frac{1}{2}}^n \right.}{\Delta x} - \frac{H_x \left _{k,l+\frac{1}{2},m+\frac{1}{2}}^n - H_x \left _{k,l-\frac{1}{2},m+\frac{1}{2}}^n \right.}{\Delta y} - J_z \left _{k,l,m+\frac{1}{2}}^n \right. \end{aligned}$	
0707.4470_EQ0020	012	0.761 K +0	$\int_{\mathrm{d}\sigma} \mathrm{d} \alpha = \int_{\partial \sigma} \alpha.$	$\int_{\sigma} \mathrm{d} \alpha = \int_{\partial \sigma} \alpha.$	$\int_{\sigma} \mathrm{d} \alpha = \int_{\partial \sigma} \alpha.$
0707.4470_EQ0009	007	0.764 N +0	$S[A] = \int_X \mathscr{L}.$	$S[A] = \int_X \mathscr{L}.$	$S[A] = \int_X \mathscr{L}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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0707.4470_EQ0030	018	■ 0.881 ● W +0	$\begin{aligned} y \ z \ t \ \text{\text{-face: }} & \& -\left(\right. \\ & \left. \left. E_y\right _{k,l+\frac{1}{2},m+1} - E_y\right _{k,l+\frac{1}{2},m} \Big) \Delta y \Delta t \\ & + \left(E_z\right _{k,l+1,m+\frac{1}{2}} - E_z\right _{k,l,m+\frac{1}{2}} \Big) \Delta z \Delta t \\ & + \left(B_x\right _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1} - B_x\right _{k,l+\frac{1}{2},m+\frac{1}{2}}^n \Big) \Delta y \Delta z \end{aligned}$	$\begin{aligned} yzt\text{-face:} & - \left(E_y\right _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}} - E_y\right _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}} \Big) \Delta y \Delta t \\ & + \left(E_z\right _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z\right _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}} \Big) \Delta z \Delta t \\ & + \left(B_x\right _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1} - B_x\right _{k,l+\frac{1}{2},m+\frac{1}{2}}^n \Big) \Delta y \Delta z \end{aligned}$	$\begin{aligned} yzt\text{-face:} & - \left(E_y\right _{k,l+\frac{1}{2},m+1}^{n+\frac{1}{2}} - E_y\right _{k,l+\frac{1}{2},m}^{n+\frac{1}{2}} \Big) \Delta y \Delta t \\ & + \left(E_z\right _{k,l+1,m+\frac{1}{2}}^{n+\frac{1}{2}} - E_z\right _{k,l,m+\frac{1}{2}}^{n+\frac{1}{2}} \Big) \Delta z \Delta t \\ & + \left(B_x\right _{k,l+\frac{1}{2},m+\frac{1}{2}}^{n+1} - B_x\right _{k,l+\frac{1}{2},m+\frac{1}{2}}^n \Big) \Delta y \Delta z \end{aligned}$
0707.4470_EQ0058 (5.3)	032	■ 0.887 ● N +2	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & = \int_X \left(\mathbf{d}_t \alpha \wedge D - \mathbf{d}_s \alpha \wedge H \wedge \mathbf{d} t + \alpha \wedge J \wedge \mathbf{d} t \right) \\ & = \int_X \alpha \wedge \left(\mathbf{d}_t D - \mathbf{d}_s H \wedge \mathbf{d} t + J \wedge \mathbf{d} t \right) \end{aligned}$	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & = \int_X \left(\mathbf{d}_t \alpha \wedge D - \mathbf{d}_s \alpha \wedge H \wedge \mathbf{d} t + \alpha \wedge J \wedge \mathbf{d} t \right) \\ & = \int_X \alpha \wedge \left(\mathbf{d}_t D - \mathbf{d}_s H \wedge \mathbf{d} t + J \wedge \mathbf{d} t \right) \end{aligned}$	$\begin{aligned} \mathbf{d} S[A] \cdot \alpha & = \int_X \left(\mathbf{d}_t \alpha \wedge D - \mathbf{d}_s \alpha \wedge H \wedge \mathbf{d} t + \alpha \wedge J \wedge \mathbf{d} t \right) \\ & = \int_X \alpha \wedge \left(\mathbf{d}_t D - \mathbf{d}_s H \wedge \mathbf{d} t + J \wedge \mathbf{d} t \right). \end{aligned}$
0707.4470_EQ0059	032	■ 0.892 ● K +0	$\mathbf{d} S[A] \cdot \alpha = \int_{\Sigma} \alpha \wedge D \Big _{t_0}^{t_f},$	$\mathbf{d} S[A] \cdot \alpha = \int_{\Sigma} \alpha \wedge D \Big _{t_0}^{t_f},$	$\mathbf{d} S[A] \cdot \alpha = \int_{\Sigma} \alpha \wedge D \Big _{t_0}^{t_f},$
0707.4470_EQ0016 (2.5)	008	■ 0.911 ● N -1	$\begin{aligned} \mathbf{d} S[A, F, G] \cdot (\alpha, \phi, \gamma) & = \int_X [-\phi \wedge *F + \alpha \wedge \mathcal{J} + (\phi - \mathbf{d}\alpha) \wedge G + (F - \mathbf{d}A) \wedge \gamma] \\ & = \int_X [\alpha \wedge (\mathcal{J} - \mathbf{d}G) + \phi \wedge (G - *F) + (F - \mathbf{d}A) \wedge \gamma] \end{aligned}$	$\begin{aligned} \mathbf{d} S[A, F, G] \cdot (\alpha, \phi, \gamma) & = \int_X [-\phi \wedge *F + \alpha \wedge \mathcal{J} + (\phi - \mathbf{d}\alpha) \wedge G + (F - \mathbf{d}A) \wedge \gamma] \\ & = \int_X [\alpha \wedge (\mathcal{J} - \mathbf{d}G) + \phi \wedge (G - *F) + (F - \mathbf{d}A) \wedge \gamma] \end{aligned}$	$\begin{aligned} \mathbf{d} S[A, F, G] \cdot (\alpha, \phi, \gamma) & = \int_X [-\phi \wedge *F + \alpha \wedge \mathcal{J} + (\phi - \mathbf{d}\alpha) \wedge G + (F - \mathbf{d}A) \wedge \gamma] \\ & = \int_X [\alpha \wedge (\mathcal{J} - \mathbf{d}G) + \phi \wedge (G - *F) + (F - \mathbf{d}A) \wedge \gamma]. \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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0707.4470_E00028	018	0.937 W +0	$\begin{aligned} & x \, y \, t \, \text{-face} : \} \& -\left(\right. \\ & \left. \left. \left. E_x \right _{k+\frac{1}{2}, l+1, m} - E_x \right _{k+\frac{1}{2}, l, m} \right)^{n+\frac{1}{2}} \Delta x \Delta t \\ & + \left(E_y \right _{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ & + \left(B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1} - B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \Delta x \Delta y \end{aligned}$	$\begin{aligned} & xyt\text{-face} : \quad - \left(E_x \right _{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}} - E_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \Delta x \Delta t \\ & + \left(E_y \right _{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ & + \left(B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1} - B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \Delta x \Delta y \end{aligned}$	$\begin{aligned} & xyt\text{-face} : - \left(E_x \right _{k+\frac{1}{2}, l+1, m}^{n+\frac{1}{2}} - E_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} \Delta x \Delta t \\ & + \left(E_y \right _{k+1, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - E_y \right _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} \Delta y \Delta t \\ & + \left(B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^{n+1} - B_z \right _{k+\frac{1}{2}, l+\frac{1}{2}, m}^n \Delta x \Delta y \end{aligned}$
0707.4470_E00036 (4.2)	020	0.951 W +1	$\begin{aligned} & \& \frac{\left(D_x \right _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} - D_x \right _{k-\frac{1}{2}, l, m}^{n+\frac{1}{2}}}{\Delta x} + \frac{D_y \left _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - D_y \right _{k, l-\frac{1}{2}, m}^{n+\frac{1}{2}}}{\Delta y} \\ & + \frac{D_z \left _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - D_z \right _{k, l, m-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} = \rho \right _{k, l, m}^{n+\frac{1}{2}} . \end{aligned}$	$\begin{aligned} & \frac{D_x \left _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} - D_x \right _{k-\frac{1}{2}, l, m}^{n+\frac{1}{2}}}{\Delta x} + \frac{D_y \left _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - D_y \right _{k, l-\frac{1}{2}, m}^{n+\frac{1}{2}}}{\Delta y} \\ & + \frac{D_z \left _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - D_z \right _{k, l, m-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} = \rho \right _{k, l, m}^{n+\frac{1}{2}} . \end{aligned}$	$\begin{aligned} & \frac{D_x \left _{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}} - D_x \right _{k-\frac{1}{2}, l, m}^{n+\frac{1}{2}}}{\Delta x} + \frac{D_y \left _{k, l+\frac{1}{2}, m}^{n+\frac{1}{2}} - D_y \right _{k, l-\frac{1}{2}, m}^{n+\frac{1}{2}}}{\Delta y} \\ & + \frac{D_z \left _{k, l, m+\frac{1}{2}}^{n+\frac{1}{2}} - D_z \right _{k, l, m-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} = \rho \right _{k, l, m}^{n+\frac{1}{2}} . \end{aligned}$
0707.4470_E00023	013	0.955 N +0	$\begin{aligned} & (\alpha, \beta) = \sum_{\sigma^k} \kappa(\sigma) \binom{n}{k} \frac{ \mathrm{CH}(\sigma, * \sigma) }{ \sigma ^2} \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \\ & = \sum_{\sigma^k} \kappa(\sigma) \frac{ \sigma }{ \sigma } \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \end{aligned}$	$\begin{aligned} & (\alpha, \beta) = \sum_{\sigma^k} \kappa(\sigma) \binom{n}{k} \frac{ \mathrm{CH}(\sigma, * \sigma) }{ \sigma ^2} \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \\ & = \sum_{\sigma^k} \kappa(\sigma) \frac{ \sigma }{ \sigma } \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \end{aligned}$	$\begin{aligned} & (\alpha, \beta) = \sum_{\sigma^k} \kappa(\sigma) \binom{n}{k} \frac{ \mathrm{CH}(\sigma, * \sigma) }{ \sigma ^2} \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \\ & = \sum_{\sigma^k} \kappa(\sigma) \frac{ \sigma }{ \sigma } \langle \alpha, \sigma \rangle \langle \beta, \sigma \rangle \end{aligned}$
0707.4470_E00008	007	0.963 N -2	$\mathscr{L} = -\frac{1}{2} \, \mathrm{d}A \wedge * \, \mathrm{d}A + A \wedge \mathscr{J} ,$	$\mathscr{L} = -\frac{1}{2} \, \mathrm{d}A \wedge * \mathrm{d}A + A \wedge \mathscr{J} ,$	$\mathscr{L} = -\frac{1}{2} \mathrm{d}A \wedge * \mathrm{d}A + A \wedge \mathscr{J} ,$
0707.4470_E00012 (2.3)	008	0.982 N +0	$\begin{aligned} & \& \mathrm{d} F = 0 \quad \& \mathrm{d} G = \mathscr{J} \end{aligned}$	$\begin{aligned} & \mathrm{d} F = 0 \\ & \mathrm{d} G = \mathscr{J} \end{aligned}$	$\begin{aligned} & \mathrm{d} F = 0 \\ & \mathrm{d} G = \mathscr{J} \end{aligned}$
0707.4470_E00060	032	0.982 K +0	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_{\Sigma} \mathrm{d}_s f \wedge D \Big _{t_0}^{t_f} = - \int_{\Sigma} f \wedge \mathrm{d}_s D \Big _{t_0}^{t_f}$	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_{\Sigma} \mathrm{d}_s f \wedge D \Big _{t_0}^{t_f} = - \int_{\Sigma} f \wedge \mathrm{d}_s D \Big _{t_0}^{t_f}$	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_{\Sigma} \mathrm{d}_s f \wedge D \Big _{t_0}^{t_f} = - \int_{\Sigma} f \wedge \mathrm{d}_s D \Big _{t_0}^{t_f}$
Conf. MathPix confidence: 0.937 0.951 0.955 0.963 0.982 0.982 0.5-0.9 0.937 0.951 0.955 0.963 0.982 0.982 <0.5 0.937 0.951 0.955 0.963 0.982 0.982 no value 0.937 0.951 0.955 0.963 0.982 0.982 Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_ E09046 (4.4)	024	■ 0.988 ● W +0	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{D_e^{m+1/2}-D_e^{m-1/2}}{t_e^{m+1/2}-t_e^{m-1/2}}\right)=\mathrm{d}_1^T\left(H_\sigma^n\mathbb{K}\{t_\sigma^n=t_e^m\}\right)-J_e^m,$		$\frac{D_e^{m+1/2}-D_e^{m-1/2}}{t_e^{m+1/2}-t_e^{m-1/2}}=\mathrm{d}_1^T\left(H_\sigma^n\mathbb{K}\{t_\sigma^n=t_e^m\}\right)-J_e^m,$
0707.4470_E09092	006	■ 0.996 ● N -1	$\oint_C \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_S \mathbf{B} \cdot \mathrm{d}\mathbf{A},$		$\oint_C \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \int_S \mathbf{B} \cdot \mathrm{d}\mathbf{A},$
0707.4470_ E09013 (2.4)	008	■ 0.996 ● N +0	$\begin{aligned} \nabla \times \mathbf{E} + \partial_t \mathbf{B} &= 0 \\ \nabla \cdot \mathbf{B} &= 0. \end{aligned}$		$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$ $\nabla \cdot \mathbf{B} = 0.$
0707.4470_E09091	005	■ 0.997 ● W +1	$\begin{aligned} &E=E_x\,\mathrm{d}x+E_y\,\mathrm{d}y+E_z\,\mathrm{d}z \\ &B=B_x\,\mathrm{d}y\wedge\mathrm{d}z+B_y\,\mathrm{d}z\wedge\mathrm{d}x+B_z\,\mathrm{d}x\wedge\mathrm{d}y, \end{aligned}$		$E=E_x\,\mathrm{d}x+E_y\,\mathrm{d}y+E_z\,\mathrm{d}z$ $B=B_x\,\mathrm{d}y\wedge\mathrm{d}z+B_y\,\mathrm{d}z\wedge\mathrm{d}x+B_z\,\mathrm{d}x\wedge\mathrm{d}y,$
0707.4470_E09047	029	■ 0.997 ● N -2	$\mathcal{L}_d=-\frac{1}{2}\,\mathrm{d}A\wedge*\mathrm{d}A+A\wedge\mathcal{J},$		$\mathcal{L}_d=-\frac{1}{2}\mathrm{d}A\wedge*\mathrm{d}A+A\wedge\mathcal{J},$
0707.4470_ E09015 (2.7)	008	■ 0.999 ● N -1	$S[A,F,G]=\int_X\left[-\frac{1}{2}F\wedge*F+A\wedge\mathcal{J}+(F-\mathrm{d}A)\wedge G\right].$		$S[A,F,G]=\int_X\left[-\frac{1}{2}F\wedge*F+A\wedge\mathcal{J}+(F-\mathrm{d}A)\wedge G\right].$
0707.4470_E09010	007	■ 1.000 ● N -2	$\begin{aligned} \mathbf{d}S[A]\cdot\alpha&=\frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big _{\epsilon=0}S[A+\epsilon\alpha] \\ &=\int_X(-\mathrm{d}\alpha\wedge*\mathrm{d}A+\alpha\wedge\mathcal{J}) \\ &=\int_X\alpha\wedge(-\mathrm{d}* \mathrm{d}A+\mathcal{J}), \end{aligned}$		$\mathbf{d}S[A]\cdot\alpha=\frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big _{\epsilon=0}S[A+\epsilon\alpha]$ $=\int_X(-\mathrm{d}\alpha\wedge*\mathrm{d}A+\alpha\wedge\mathcal{J})$ $=\int_X\alpha\wedge(-\mathrm{d}* \mathrm{d}A+\mathcal{J}),$
0707.4470_ E09033 (4.1)	019	■ 1.000 ● W +2	$\begin{aligned} &\frac{B_x _{k+1,l+\frac{1}{2},m+\frac{1}{2}}-B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}}{\Delta x}+\frac{B_y _{k+\frac{1}{2},l+1,m+\frac{1}{2}}-B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}}{\Delta y} \\ &+\frac{B_z _{k+\frac{1}{2},l+\frac{1}{2},m+1}-B_z _{k+\frac{1}{2},l+\frac{1}{2},m}}{\Delta z}=0. \end{aligned}$		$\frac{B_x _{k+1,l+\frac{1}{2},m+\frac{1}{2}}-B_x _{k,l+\frac{1}{2},m+\frac{1}{2}}}{\Delta x}+\frac{B_y _{k+\frac{1}{2},l+1,m+\frac{1}{2}}-B_y _{k+\frac{1}{2},l,m+\frac{1}{2}}}{\Delta y}$ $+\frac{B_z _{k+\frac{1}{2},l+\frac{1}{2},m+1}-B_z _{k+\frac{1}{2},l+\frac{1}{2},m}}{\Delta z}=0.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

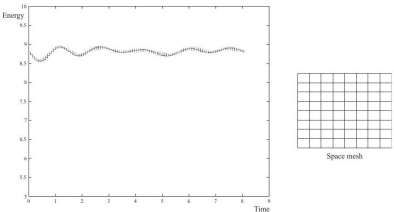
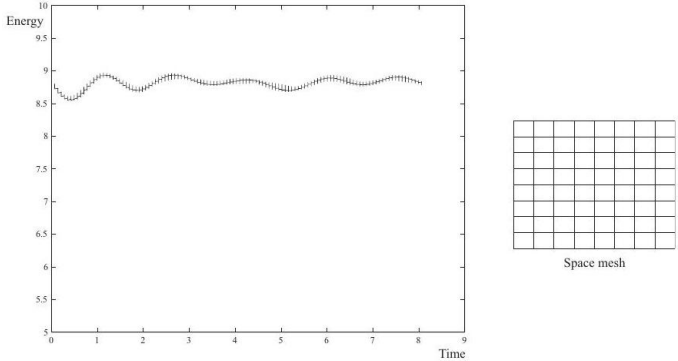
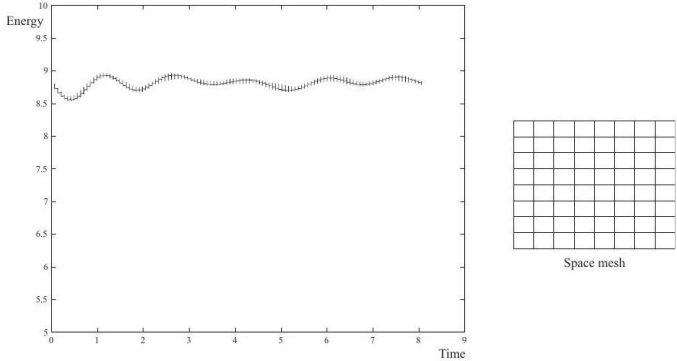
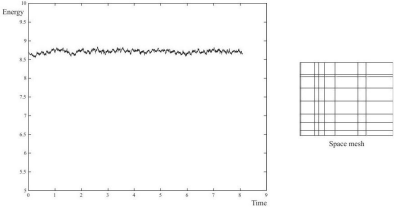
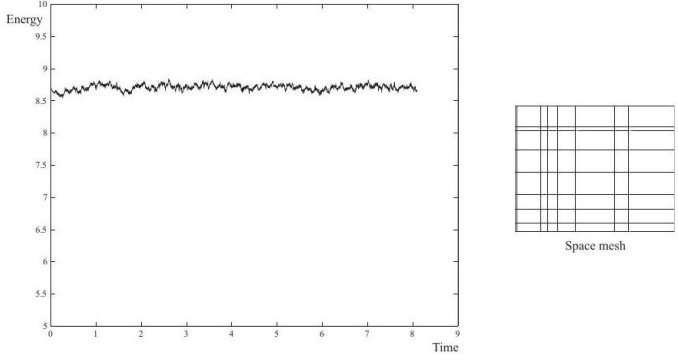
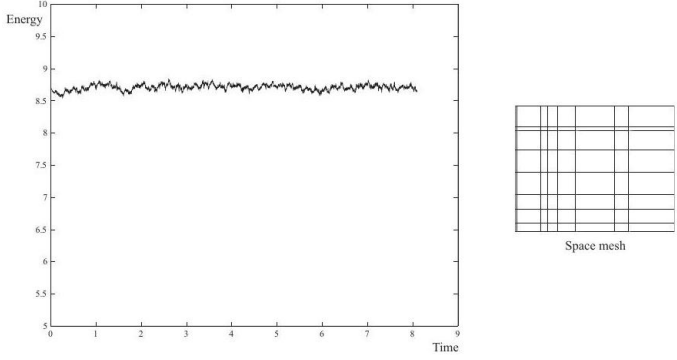
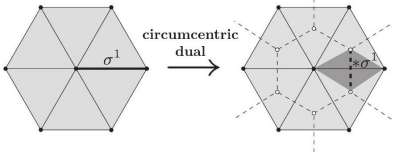
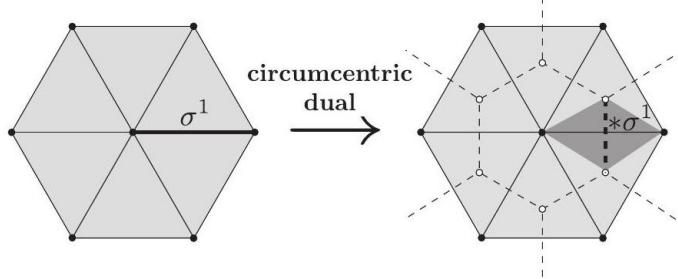
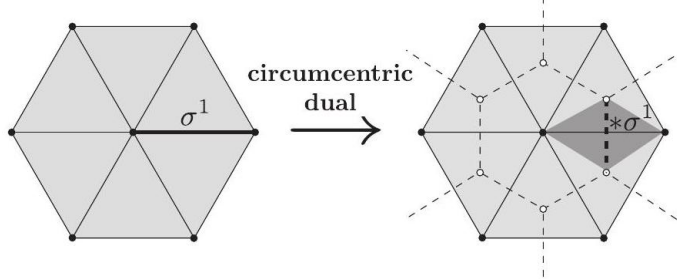
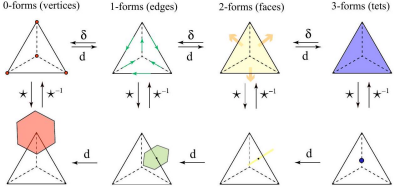
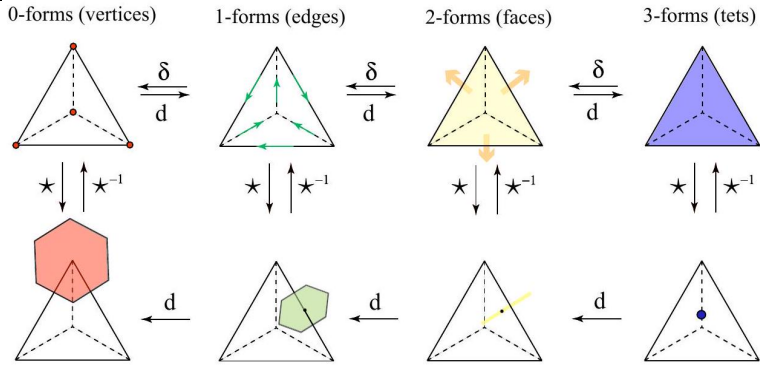
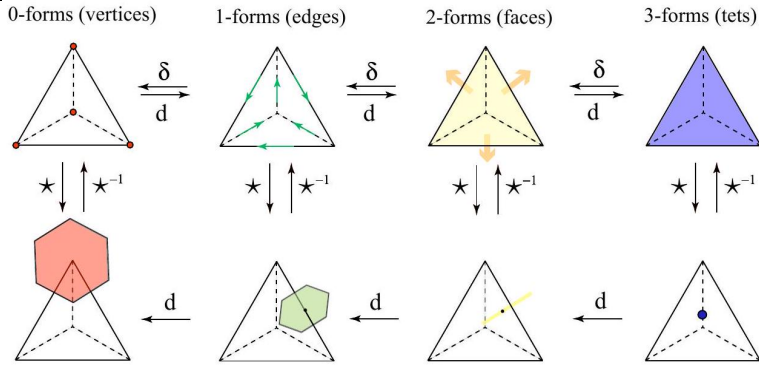
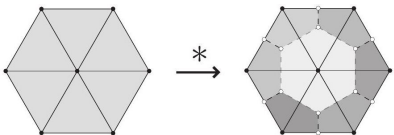
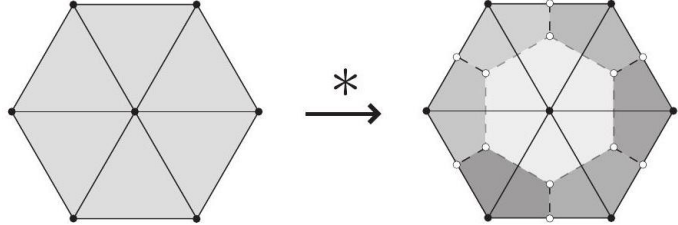
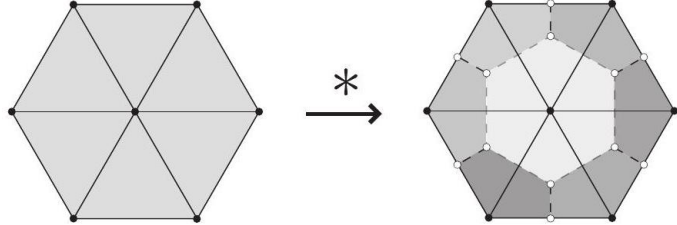
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_E00049	029	■ 1.000 ● W -2	$\mathrm{d} S_{\mathrm{d}}[A] \cdot \alpha = \langle -\mathrm{d} \alpha \wedge * \mathrm{d} A + \alpha \wedge \mathcal{J}, K \rangle = \langle \alpha \wedge (-\mathrm{d} * \mathrm{d} A + \mathcal{J}), K \rangle.$	$\mathrm{d} S_d[A] \cdot \alpha = \langle -\mathrm{d} \alpha \wedge * \mathrm{d} A + \alpha \wedge \mathcal{J}, K \rangle = \langle \alpha \wedge (-\mathrm{d} * \mathrm{d} A + \mathcal{J}), K \rangle.$	
0707.4470_E00050	030	■ 1.000 ● N -2	$\mathrm{d} S_{\mathrm{d}}[A] \cdot \alpha = \langle \alpha \wedge (-\mathrm{d} * \mathrm{d} A + \mathcal{J}), K \rangle + \langle \alpha \wedge * d A, \partial K \rangle.$	$\mathrm{d} S_d[A] \cdot \alpha = \langle \alpha \wedge (-\mathrm{d} * \mathrm{d} A + \mathcal{J}), K \rangle + \langle \alpha \wedge * d A, \partial K \rangle.$	
0707.4470_E00003	006	■ 1.000 ● N -1	$\oint_C \mathbf{H} \cdot \mathrm{d} \mathbf{l} = \frac{\mathrm{d}}{\mathrm{d} t} \int_S \mathbf{D} \cdot \mathrm{d} \mathbf{A},$	$\oint_C \mathbf{H} \cdot \mathrm{d} \mathbf{l} = \frac{\mathrm{d}}{\mathrm{d} t} \int_S \mathbf{D} \cdot \mathrm{d} \mathbf{A},$	
0707.4470_E00004	006	■ 1.000 ● K +0	$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}.$	$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}.$	
0707.4470_E00005	006	■ 1.000 ● K +0	$F = E \wedge \mathrm{d} t + B.$	$F = E \wedge \mathrm{d} t + B.$	
0707.4470_E00006	007	■ 1.000 ● K +0	$G = * F = H \wedge \mathrm{d} t - D,$	$G = * F = H \wedge \mathrm{d} t - D,$	
0707.4470_E00007	007	■ 1.000 ● K +0	$\mathcal{J} = J \wedge \mathrm{d} t - \rho.$	$\mathcal{J} = J \wedge \mathrm{d} t - \rho.$	
0707.4470_E00011 (2.1)	007	■ 1.000 ● N -1	$\mathrm{d} * \mathrm{d} A = \mathcal{J}.$	$\mathrm{d} * \mathrm{d} A = \mathcal{J}.$	
0707.4470_E00014 (2.6)	008	■ 1.000 ● K +0	$\begin{aligned} \nabla \times \mathbf{H} - \partial_t \mathbf{D} &= \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho. \end{aligned}$	$\begin{aligned} \nabla \times \mathbf{H} - \partial_t \mathbf{D} &= \mathbf{J} \\ \nabla \cdot \mathbf{D} &= \rho. \end{aligned}$	
0707.4470_E00017 (2.2)	008	■ 1.000 ● N -1	$\mathrm{d} G = \mathcal{J}, \quad G = * F, \quad F = \mathrm{d} A.$	$\mathrm{d} G = \mathcal{J}, \quad G = * F, \quad F = \mathrm{d} A.$	
0707.4470_E00018	009	■ 1.000 ● K +0	$\partial_t (\nabla \cdot \mathbf{B}) = 0, \quad \partial_t (\nabla \cdot \mathbf{D}) + \nabla \cdot \mathbf{J} = \partial_t (\nabla \cdot \mathbf{D} - \rho) = 0.$	$\partial_t (\nabla \cdot \mathbf{B}) = 0, \quad \partial_t (\nabla \cdot \mathbf{D}) + \nabla \cdot \mathbf{J} = \partial_t (\nabla \cdot \mathbf{D} - \rho) = 0.$	
0707.4470_E00019	012	■ 1.000 ● N +0	$\langle \alpha, \sigma \rangle \equiv \int_{\sigma} \alpha.$	$\langle \alpha, \sigma \rangle \equiv \int_{\sigma} \alpha.$	
0707.4470_E00021	012	■ 1.000 ● K +0	$\langle \mathrm{d} \alpha, \sigma \rangle = \langle \alpha, \partial \sigma \rangle,$	$\langle \mathrm{d} \alpha, \sigma \rangle = \langle \alpha, \partial \sigma \rangle,$	
0707.4470_E00022	013	■ 1.000 ● K +0	$\frac{1}{ * \sigma } \langle * \alpha, * \sigma \rangle = \kappa(\sigma) \frac{1}{ \sigma } \langle \alpha, \sigma \rangle,$	$\frac{1}{ * \sigma } \langle * \alpha, * \sigma \rangle = \kappa(\sigma) \frac{1}{ \sigma } \langle \alpha, \sigma \rangle,$	
0707.4470_E00024 (3.1)	016	■ 1.000 ● N +0	$(\mathrm{d} \alpha, \beta) = (\alpha, \delta \beta) + \langle \alpha \wedge * \beta, \partial K \rangle.$	$(\mathrm{d} \alpha, \beta) = (\alpha, \delta \beta) + \langle \alpha \wedge * \beta, \partial K \rangle.$	
0707.4470_E00027	018	■ 1.000 ● N +0	$\mathrm{d} F = 0, \quad \mathrm{d} G = \mathcal{J},$	$\mathrm{d} F = 0, \quad \mathrm{d} G = \mathcal{J},$	
0707.4470_E00034	019	■ 1.000 ● K +0	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B} = 0.$	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \quad \nabla \cdot \mathbf{B} = 0.$	
0707.4470_E00037	020	■ 1.000 ● K +0	$\partial_t \mathbf{D} = \nabla \times \mathbf{H} - \mathbf{J}, \quad \nabla \cdot \mathbf{D} = \rho.$	$\partial_t \mathbf{D} = \nabla \times \mathbf{H} - \mathbf{J}, \quad \nabla \cdot \mathbf{D} = \rho.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

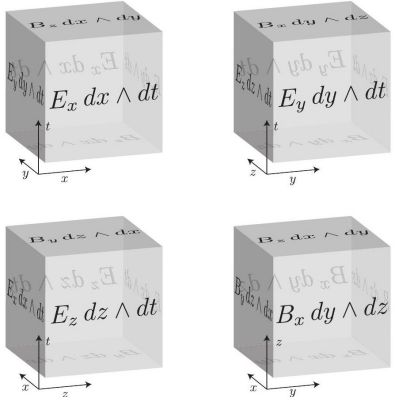
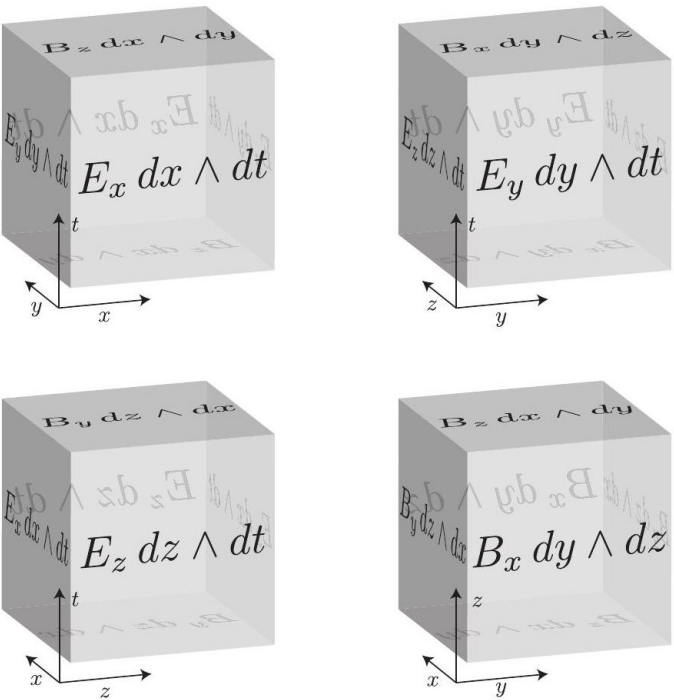
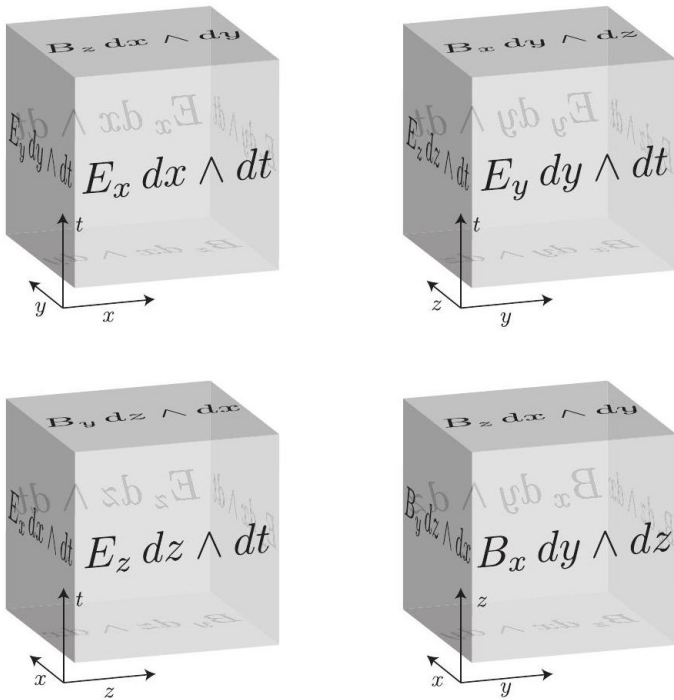
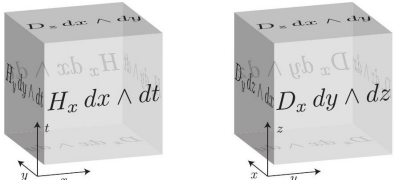
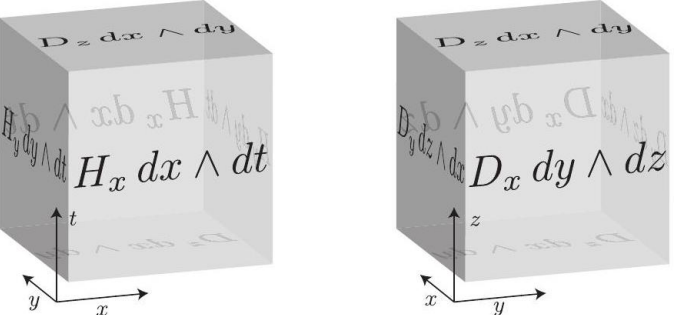
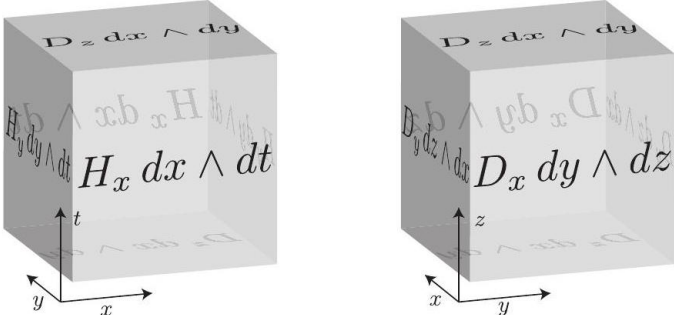
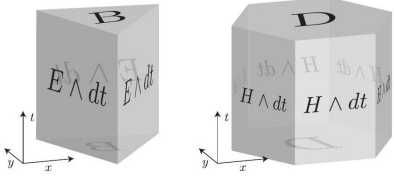
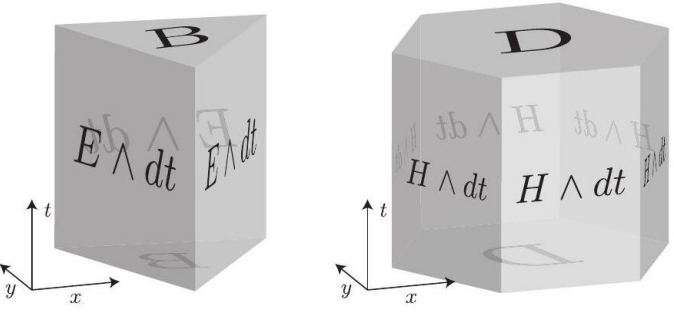
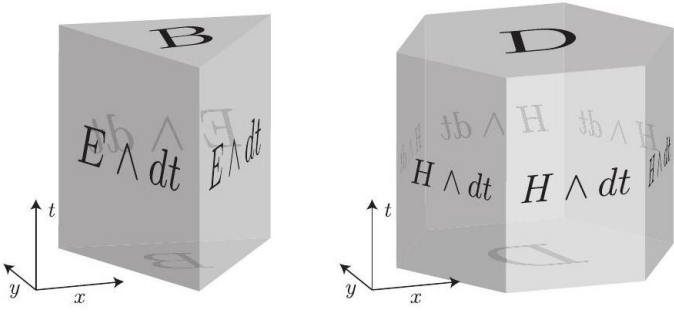
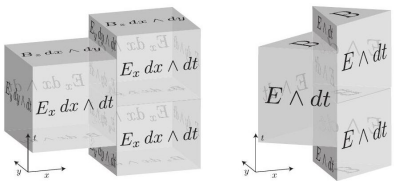
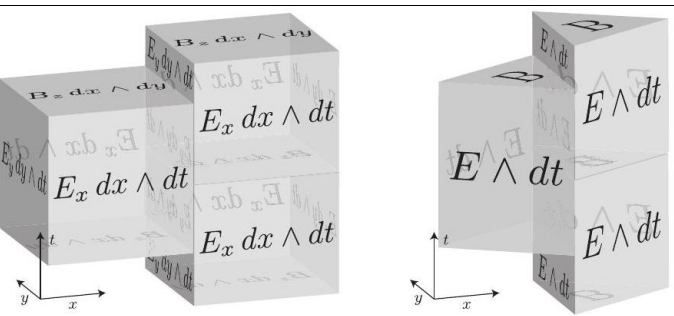
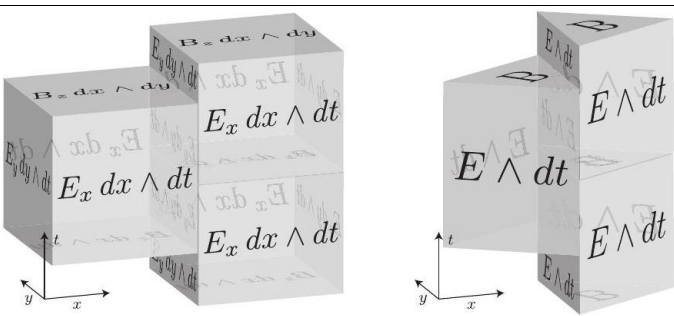
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_E00038	021	■ 1.000 ● K +0	$F=E \wedge \mathrm{d} t+B$.	$F = E \wedge \mathrm{d} t + B.$	$F = E \wedge \mathrm{d} t + B.$
0707.4470_E00039	022	■ 1.000 ● N +0	$\frac{B^{n+1}-B^n}{\Delta t}=-\mathrm{d}_1 E^{n+1/2}$.	$\frac{B^{n+1}-B^n}{\Delta t} = -\mathrm{d}_1 E^{n+1/2}.$	$\frac{B^{n+1}-B^n}{\Delta t} = -\mathrm{d}_1 E^{n+1/2}.$
0707.4470_E00040	022	■ 1.000 ● S +0	$\frac{D^{n+1/2}-D^{n-1/2}}{\Delta t}=\mathrm{d}_1^T H^n-J^n$.	$\frac{D^{n+1/2}-D^{n-1/2}}{\Delta t} = \mathrm{d}_1^T H^n - J^n.$	$\frac{D^{n+1/2}-D^{n-1/2}}{\Delta t} = \mathrm{d}_1^T H^n - J^n.$
0707.4470_E00041	023	■ 1.000 ● K +0	$\Theta_{\sigma}=\left\{t_{\sigma}^0<\cdots<t_{\sigma}^{N_{\sigma}}\right\}$.	$\Theta_{\sigma} = \left\{t_{\sigma}^0 < \cdots < t_{\sigma}^{N_{\sigma}}\right\}.$	$\Theta_{\sigma} = \left\{t_{\sigma}^0 < \cdots < t_{\sigma}^{N_{\sigma}}\right\}.$
0707.4470_E00042	023	■ 1.000 ● N +0	$\Theta_e=\bigcup_{\sigma\ni e}\Theta_{\sigma}=\left\{t_e^0\leq\cdots\leq t_e^{N_e}\right\}$.	$\Theta_e = \bigcup_{\sigma\ni e} \Theta_{\sigma} = \left\{t_e^0 \leq \cdots \leq t_e^{N_e}\right\}.$	$\Theta_e = \bigcup_{\sigma\ni e} \Theta_{\sigma} = \left\{t_e^0 \leq \cdots \leq t_e^{N_e}\right\}.$
0707.4470_E00043	023	■ 1.000 ● N +0	$\Theta_e^{\prime}=\left\{t_e^{1/2}\leq\cdots\leq t_e^{N_e-1/2}\right\},$	$\Theta_e^{\prime} = \left\{t_e^{1/2} \leq \cdots \leq t_e^{N_e-1/2}\right\},$	$\Theta_e^{\prime} = \left\{t_e^{1/2} \leq \cdots \leq t_e^{N_e-1/2}\right\},$
0707.4470_E00044	024	■ 1.000 ● N +0	$\Theta_{*\sigma}=\Theta_{\sigma},\quad \Theta_{*e}=\Theta_e.$	$\Theta_{*\sigma} = \Theta_{\sigma},\quad \Theta_{*e} = \Theta_e.$	$\Theta_{*\sigma} = \Theta_{\sigma},\quad \Theta_{*e} = \Theta_e.$
0707.4470_E00045 (4.3)	024	■ 1.000 ● W +0	$\frac{B_{\sigma}^{n+1}-B_{\sigma}^n}{t_{\sigma}^{n+1}-t_{\sigma}^n}=-\mathrm{d}_1\sum\left\{E_e^{m+1/2}:t_{\sigma}^n<t_e^{m+1/2}<t_{\sigma}^{n+1}\right\}.$	$\frac{B_{\sigma}^{n+1}-B_{\sigma}^n}{t_{\sigma}^{n+1}-t_{\sigma}^n} = -\mathrm{d}_1 \sum \left\{E_e^{m+1/2}:t_{\sigma}^n < t_e^{m+1/2} < t_{\sigma}^{n+1}\right\}.$	$\frac{B_{\sigma}^{n+1}-B_{\sigma}^n}{t_{\sigma}^{n+1}-t_{\sigma}^n} = -\mathrm{d}_1 \sum \left\{E_e^{m+1/2}:t_{\sigma}^n < t_e^{m+1/2} < t_{\sigma}^{n+1}\right\}.$
0707.4470_E00048	029	■ 1.000 ● N +0	$S_d[A]=\left\langle\mathrm{d}\mathscr{L}_d\right\rangle K$.	$S_d[A] = \left\langle\mathscr{L}_d, K\right\rangle.$	$S_d[A] = \left\langle\mathscr{L}_d, K\right\rangle.$
0707.4470_E00051 (5.1)	030	■ 1.000 ● N -1	$\mathbf{d}S_d(A)\cdot\alpha=\left\langle\alpha\wedge*\mathrm{d}A,\partial K\right\rangle$	$\mathbf{d}S_d(A)\cdot\alpha = \left\langle\alpha\wedge*\mathrm{d}A,\partial K\right\rangle$	$\mathbf{d}S_d(A)\cdot\alpha = \left\langle\alpha\wedge*\mathrm{d}A,\partial K\right\rangle$
0707.4470_E00052	030	■ 1.000 ● N -1	$\theta_{\mathscr{L}_d}\cdot\alpha=\alpha\wedge*\mathrm{d}A.$	$\theta_{\mathscr{L}_d}\cdot\alpha = \alpha\wedge*\mathrm{d}A.$	$\theta_{\mathscr{L}_d}\cdot\alpha = \alpha\wedge*\mathrm{d}A.$
0707.4470_E00053	030	■ 1.000 ● N +0	$\mathbf{d}^2S_d[A]\cdot\alpha\cdot\beta=\left\langle\mathbf{d}\theta\cdot\alpha\cdot\beta,\partial K\right\rangle.$	$\mathbf{d}^2S_d[A]\cdot\alpha\cdot\beta = \left\langle\mathbf{d}\theta\cdot\alpha\cdot\beta,\partial K\right\rangle.$	$\mathbf{d}^2S_d[A]\cdot\alpha\cdot\beta = \left\langle\mathbf{d}\theta\cdot\alpha\cdot\beta,\partial K\right\rangle.$
0707.4470_E00054 (5.2)	030	■ 1.000 ● K +0	$\left\langle\omega_{\mathscr{L}_d}\cdot\alpha\cdot\beta,\partial K\right\rangle=0$	$\left\langle\omega_{\mathscr{L}_d}\cdot\alpha\cdot\beta,\partial K\right\rangle = 0$	$\left\langle\omega_{\mathscr{L}_d}\cdot\alpha\cdot\beta,\partial K\right\rangle = 0$
0707.4470_E00055	031	■ 1.000 ● N +0	$A=A_x\mathrm{d}x+A_y\mathrm{d}y+A_z\mathrm{d}z$.	$A = A_x \mathrm{d} x + A_y \mathrm{d} y + A_z \mathrm{d} z.$	$A = A_x \mathrm{d} x + A_y \mathrm{d} y + A_z \mathrm{d} z.$
0707.4470_E00056	032	■ 1.000 ● K +0	$\mathrm{d}_tA=E\wedge\mathrm{d}t,\quad \mathrm{d}_sA=B.$	$\mathrm{d}_tA = E \wedge \mathrm{d} t,\quad \mathrm{d}_sA = B.$	$\mathrm{d}_tA = E \wedge \mathrm{d} t,\quad \mathrm{d}_sA = B.$
0707.4470_E00057	032	■ 1.000 ● N +0	$\begin{aligned} \mathscr{L} &= -\frac{1}{2}\left(\mathrm{d}_tA+\mathrm{d}_sA\right)\wedge*\left(\mathrm{d}_tA+\mathrm{d}_sA\right)+A\wedge\mathscr{J} \\ &= -\frac{1}{2}\left(\mathrm{d}_tA\wedge*\mathrm{d}_tA+\mathrm{d}_sA\wedge*\mathrm{d}_sA\right)+A\wedge J\wedge\mathrm{d}t \end{aligned}$	$\mathscr{L} = -\frac{1}{2} \left(\mathrm{d}_tA + \mathrm{d}_sA \right) \wedge * \left(\mathrm{d}_tA + \mathrm{d}_sA \right) + A \wedge \mathscr{J}$ $= -\frac{1}{2} \left(\mathrm{d}_tA \wedge * \mathrm{d}_tA + \mathrm{d}_sA \wedge * \mathrm{d}_sA \right) + A \wedge J \wedge \mathrm{d} t$	$\mathscr{L} = -\frac{1}{2} \left(\mathrm{d}_tA + \mathrm{d}_sA \right) \wedge * \left(\mathrm{d}_tA + \mathrm{d}_sA \right) + A \wedge \mathscr{J}$ $= -\frac{1}{2} \left(\mathrm{d}_tA \wedge * \mathrm{d}_tA + \mathrm{d}_sA \wedge * \mathrm{d}_sA \right) + A \wedge J \wedge \mathrm{d} t$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

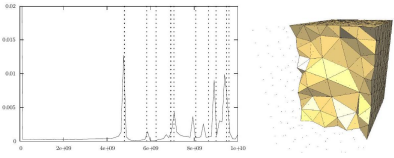
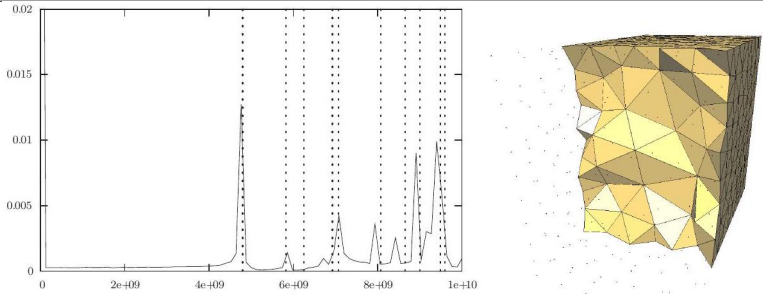
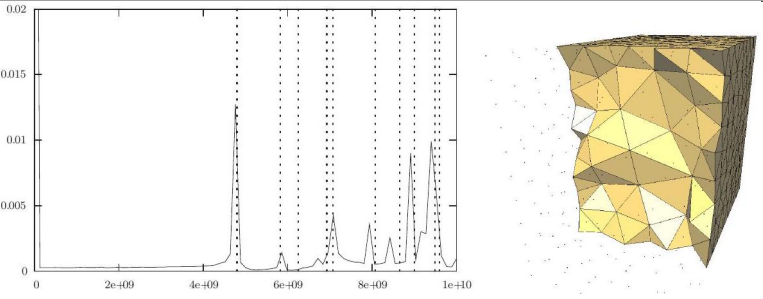
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0707.4470_EQ0061	032	■ 1.000 ● K +0	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_t \rho = - \int_\Sigma f \wedge \rho \Big _{t_0}^{t_f} .$	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_t \rho = - \int_\Sigma f \wedge \rho \Big _{t_0}^{t_f} .$	$\mathbf{d} S[A] \cdot \mathrm{d}_s f = \int_X \mathrm{d}_s f \wedge J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_s J \wedge \mathrm{d} t = - \int_X f \wedge \mathrm{d}_t \rho = - \int_\Sigma f \wedge \rho \Big _{t_0}^{t_f} .$
0707.4470_EQ0062	033	■ 1.000 ● K +0	$\left. \left. \left(\mathrm{d}_s D - \rho \right) \right _{t_0} \right _{t_f} = 0 .$	$\left(\mathrm{d}_s D - \rho \right) \Big _{t_0}^{t_f} = 0 .$	$\left(\mathrm{d}_s D - \rho \right) \Big _{t_0}^{t_f} = 0 .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
6707.4470_PIC_0001	026	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-26_475_883_406_621.jpg			
6707.4470_PIC_0002	026	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-26_466_879_1235_621.jpg			
6707.4470_DIA_0001	012	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-12_376_885_406_619.jpg			
6707.4470_DIA_0002	014	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-14_606_1263_397_430.jpg			
6707.4470_DIA_0003	015	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-15_303_885_400_619.jpg			

6707.4470_DIA_0004	017	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-17_917_894_395_614.jpg			
6707.4470_DIA_0005	021	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-21_430_887_395_617.jpg			
6707.4470_DIA_0006	022	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-22_406_887_397_617.jpg			
6707.4470_DIA_0007	023	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-23_406_889_397_617.jpg			
6707.4470_DIA_0008	025	— (dup)	no image for this region (11 in zip)	(no tex.zip crop for this row)	(image not on disk — pass -texzip or download the CDN crop)	(image not on disk — pass -texzip or download the CDN crop)

6707.4470_DIA_0009	027	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-27_455_1192_417_468.jpg			
6707.4470_DIA_0010	031	— (dup)	ec315bb1-8f7a-4d72-ac8d-58178b1ab891-31_604_634_400_743.jpg	